Riley Decker

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Research Interests

Applied topology; (multiparameter) persistence; sheaf theory; network science; representation-theoretic aspects of TDA

Education

Ph.D. in Mathematics — Expected Graduation: May 2027
University at Albany, SUNY — Albany, NY
Advisor: Michael Lesnick

B.A. in Economics — December 2016
University of Texas — Austin, TX

Papers

1. **Counting barcodes with the same Betti curve**, w/ Henry Ashley, Håvard Bakke Bjerkevik, Justin Curry, and Robert Green. Accepted to *La Matematica*, pending minor revisions. https://arxiv.org/abs/2602.09011
2. **Realization of ℓ^1-distances on multiparameter persistence modules and merge trees**, w/ Michael Lesnick. In preparation.
3. **Vietoris-Rips complexes of affine Coxeter groups**, w/ Matthew Zaremsky. In preparation.

Work Experience

- **Research Intern** at Georgia Tech Research Institute — Summer 2024

  Developed sheaf-theoretic methods to detect anomalous activity in network data using an autoencoder. Supervised by Branden Stone.

Talks

- **The structure of merge tree presentations**, Union College Mathematics Conference, September 2026.
- **Realizability of p-presentation distances**, UAlbany Algebra and Topology Seminar, October 2025.

- **Realizability of p-presentation distances**, ComPer 2025: Workshop on Computational Topology, UAlbany, October 2025.
- **Counting barcodes with the same Betti curve**, AMS Southeastern Sectional Meeting, Special Session on Topological Data Analysis, Clemson University, March 2025. Invited.
- **Sheaf analysis of network data**, JMM 2025, AMS Special Session on Topological Machine Learning, January 2025. Invited.
- **Counting barcodes with the same Betti curve**, AMS Eastern Sectional Meeting, Special Session on Applied and Computational Topology, UAlbany, October 2024.
- **Presentation distances for persistence modules**, Applied Topology in Albany Seminar, October 2023.

Teaching Experience

University at Albany, SUNY

Instructor of Record

- Contemporary Topics in Mathematics (Spring 2026)
- Calculus II (Summer 2025, Fall 2025)
- Calculus I (Spring 2024, Fall 2024, Spring 2025)
- Precalculus (Spring 2023, Fall 2023)

Teaching Assistant

- Introductory Statistics (Fall 2021, Spring 2022, Fall 2022)

Service

- Organizer, Applied Topology in Albany Seminar (Spring 2026)
- Referee for the International Symposium on Computational Geometry

Technical Skills

Programming: Python, SQL
Languages: English (native), French (reading knowledge), Spanish (reading knowledge)